TECHNICAL REPORT

VoxTrace

Contrastive speech analytics and temporal flaw grounding



EVIDENCE FIRST

VoxTrace compares an effective reference delivery with a target recording of the same script, locates deviations to exact time windows, and explains every finding with a measured value, a transparent rule, and a practice action.

AUTHOR	EVENT	DATE
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01 / METHOD

From audio to an auditable finding

WAV + transcript **VALIDATE** Feature windows **COMPARE** Timestamped findings

Problem and contribution

Speech feedback often says *too fast*, *flat*, or *quiet* without showing where the failure occurred or how the judgement was produced. VoxTrace contributes a licence-aware same-script dataset workflow, bounded acoustic feature windows, optional phoneme forced alignment, reference-derived thresholds, normalized temporal comparison, deterministic explanations, and speaker-held-out evaluation utilities.

Measured feature families

Family	Measurements	Source and interpretation
Amplitude	dBFS energy; conservative silence duration	Raw WAV samples; no inferred loudness labels
Pitch	Mean F0; semitone variation	Voiced 50 ms frames only; absent when no candidate is reliable
Spectrum	Centroid; 85% rolloff; flatness; ZCR; 13 cepstral means	Hann-windowed FFT and log-power cosine transform
Timing	WPM; word-boundary pauses	Reviewed or forced-aligned word timestamps only
Labels	Type; start/end; severity; review state	Human-reviewed manifest intervals

Feature equations

Energy is **20 log10(RMS / full scale)**. Spectral centroid is the power-weighted mean frequency; rolloff is the frequency below 85% of spectral power; flatness is the geometric-to-arithmetic magnitude ratio. Severity is a bounded relative deviation: **min(1, |x - threshold| / max(|threshold|, 1))**. It is explicitly not a probability or model confidence.

Trust boundaries

The API bounds file size, duration, transcript length, window count, and numeric values. Baseline uploads accept mono or stereo 16-bit PCM WAV and are deleted after extraction. Filenames and transcripts never become commands. Alignment is disabled by default because model loading is resource intensive.

02 / DATA

Contrastive same-script dataset design

VoxTrace-Contrastive-1 pairs every script's effective reference with controlled variants: near-perfect, rushed, excessive-pause, low-energy, and mixed-flaw. The generator creates reviewable variants from a licensed reference WAV while preserving the transcript. Generated intervals remain **pending** until human review; the system never reports generated labels as ground truth.

Manifest field	Purpose
recording_id / script_id / speaker_id	Stable provenance and pseudonymous speaker grouping
audio_path / sample_rate / duration	Reproducible decoding and window boundaries
source / licence / consent_record	Redistribution and participant-permission audit trail
transcript / variant	Same-script pairing and stress condition
labels[]	Typed temporal intervals with severity and review state

Leakage prevention

Dataset splits operate by speaker, never by recording. For train and test speaker sets, VoxTrace enforces **S(train) intersection S(test) = empty**. This prevents the same voice from appearing in threshold calibration and evaluation.

Normalized temporal comparison

Deliberate pacing changes total duration, so a target window midpoint **t** in target duration **T** maps to expected reference time **(t/T)R** for reference duration **R**. The nearest reference midpoint supplies feature deltas. Word-level forced alignment is used when exact lexical timing is required.

REFERENCE	TARGET	OUTPUT
Robust rate band; pause ceiling; pitch, energy and clarity floors	Measured one-second windows mapped by normalized midpoint	Observed value + baseline + delta + exact interval

Forced alignment

The optional worker uses a language-specific WhisperX phoneme model with the fixed transcript. Returned word times must be finite, ordered, and non-overlapping; unaligned tokens are not fabricated. Timing provenance is included in the response. The lightweight installation keeps this worker isolated from the default web process.

03 / EVIDENCE

Scoring, explanations, and evaluation

Deterministic scoring

Pace and pauses, prosody, energy, and clarity use explicit weights. A dimension is omitted when its required measurement is missing, and the overall score is renormalized only across available dimensions. Every finding binds a type, exact start/end, bounded severity, evidence sentence with units and threshold, and a concrete practice action. No generative model participates in runtime scoring.

Dashboard evidence chain

Panel	Question answered
Run provenance + completeness	Which measurements exist, and which path produced them?
Paired reference/target timeline	Where and how does the delivery diverge from the same-script baseline?
Temporal findings	What exact interval crossed which documented rule?
Practice priority	Which action addresses the most severe measured deviation first?

The hosted dashboard also has a Web Audio fallback. If the Python API is unavailable, compatible audio is decoded and measured locally in the visitor's browser; the interface states that transcript-dependent rate and clarity remain unavailable. This preserves a functional demo without pretending missing measurements exist.

Held-out evaluation

Predicted and reviewed intervals are matched one-to-one within category. Temporal IoU is intersection duration divided by union duration, with a default 0.5 match threshold. Reports contain per-class true positives, false positives, false negatives, precision, recall, F1, macro F1, and mean matched IoU.

Evidence item	Current state
Temporal evaluator	Implemented and automated-test covered
Speaker-held-out splitter	Implemented and leakage-tested
Final licensed recordings	Requires team recording and consent step
Reviewed labels	Generated labels remain pending until reviewed
Measured final metrics	Published only after final dataset collection

Reproducibility

The repository includes Docker configuration, dependency ranges, deterministic CLIs, dataset and evaluation documentation, AI-assistance disclosure, 24 passing backend tests, and a passing TypeScript/Vite production build.

04 / DELIVERY

Responsible use and reproducibility

Responsible-use limits

VoxTrace is an educational feedback prototype, not a medical, psychological, employment, or authoritative judging system. Acoustic delivery does not fully measure reasoning quality, rhetorical content, accent, culture, or audience context. Results depend on microphone, room, noise, transcript, and alignment quality. Pitch estimation is conservative; phoneme alignment may degrade for noise, code-switching, and unsupported languages.

Speech can contain biometric and personal information. The prototype avoids persistence. A production service must add authentication, ownership checks, encryption, retention controls, deletion, structured audit logs, distributed rate limiting, and a documented model-update process before storing recordings.

Submission evidence

Deliverable	Evidence
Source	github.com/abhijeetnardele24-hash/-voxtrace
Tests	24 backend tests; frontend TypeScript and production build
Dashboard	Dual upload, same-script comparison, browser-local fallback
Dataset workflow	Generator, validator, card, manifests, leakage-safe split
Report	This five-page PDF plus editable LaTeX source

AI assistance disclosure

ByteAsk/ChatGPT and OpenAI Codex assisted planning, architecture, code and test development, audit, documentation, and interface exploration. Human contributors remain responsible for code review, licences, consent, measurements, and submission claims. AI assistance is not used as runtime evidence.

Conclusion

> **VoxTrace reframes speech coaching as an auditable temporal measurement problem. Its strongest property is the evidence chain from licensed paired data, to bounded features, to reference deltas, to exact intervals, to explanations that a speaker or judge can inspect and reproduce.**

Repository: github.com/abhijeetnardele24-hash/-voxtrace